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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

4285496343

MATHEMATICS (SYLLABUS D) **4024/12**

Paper 1 **October/November 2012**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

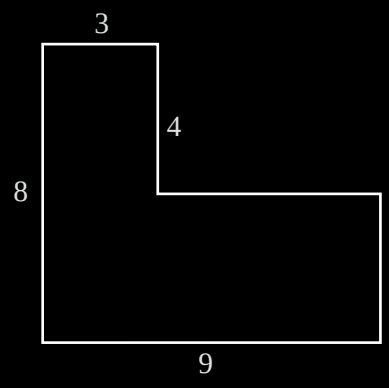
The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

DC (NH/JG) 51680/2
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Answer ..... cm [1]

(b) Square pieces, each of side 2 cm, are cut from this card.

Find the greatest number of squares that can be obtained.

Answer .....[1]

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4 $f(x) = 5 + 3x$

(a) Evaluate $f\left(-\frac{1}{2}\right)$.

Answer .....[1]

(b) Find $f^{-1}(x)$.

Answer $f^{-1}(x) =$ .....[1]

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Number of goals scored	0	1	2	3	4
Number of teams	x	1	5	4	2

(a) If the mode is 0, find the smallest possible value of x.

Answer $x =$ .....[1]

(b) If the median is 1, find the value of x.

Answer $x =$ .....[1]

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10 **(a)** Express 180 as the product of its prime factors.

Answer .....[1]

(b) $\sqrt{180}$ can be expressed in the form $p\sqrt{q}$, where p and q are integers.

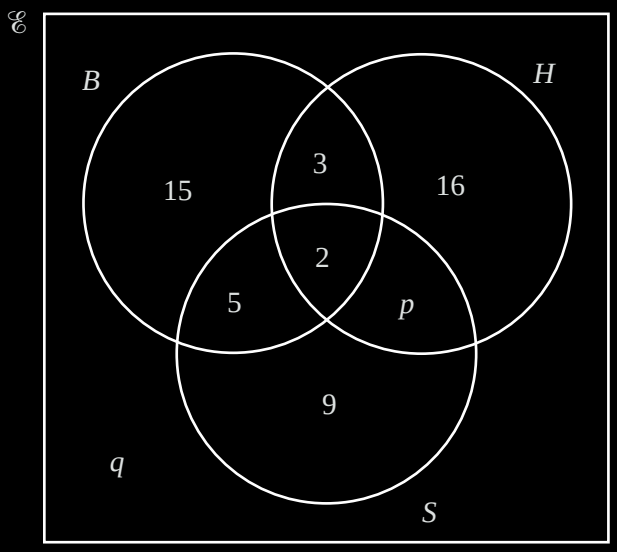
Find the smallest value of $p + q$.

Answer .....[1]

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(a) Find the values of p and q.

Answer $p =$ .....

$q =$ .....[1]

(b) Find $\mathrm{n}(H')$.

Answer .....[1]

(c) Find $\mathrm{n}((B \cup H) \cap S')$.

Answer .....[1]

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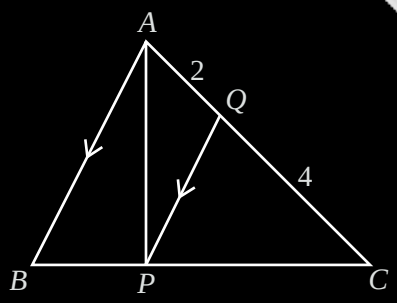
First card	A								
Second card	B								
Sum	5								

(c) What is the probability that the sum is 7?

Answer ..... [1]

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Find the area of

(a) triangle *AQP*,

Answer ..... cm^2 [1]

(b) triangle *ABC*,

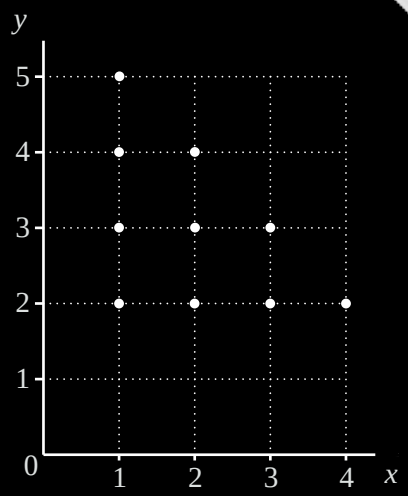
Answer ..... cm^2 [1]

(c) triangle *ABP*.

Answer ..... cm^2 [1]

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(a) For these 10 points find

(i) the maximum value of $k - h$,

Answer .....[1]

(ii) the value of k, for the point that lies on the line $y = \frac{1}{2}x$.

Answer $k =$ .....[1]

(b) The coordinates of the 10 points satisfy the inequalities

$$h \geqslant a, \qquad k \geqslant b, \qquad h + k \leqslant c.$$

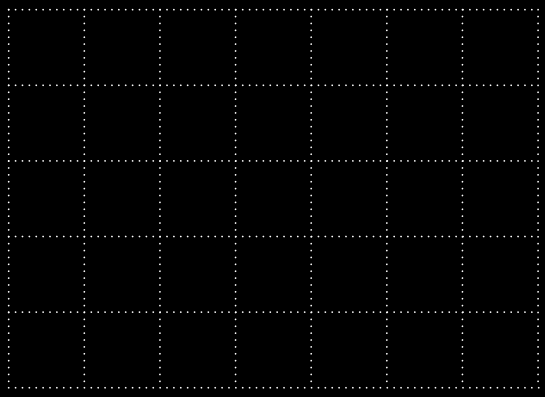
Write down the values of a, b and c.

Answer $a =$ .....

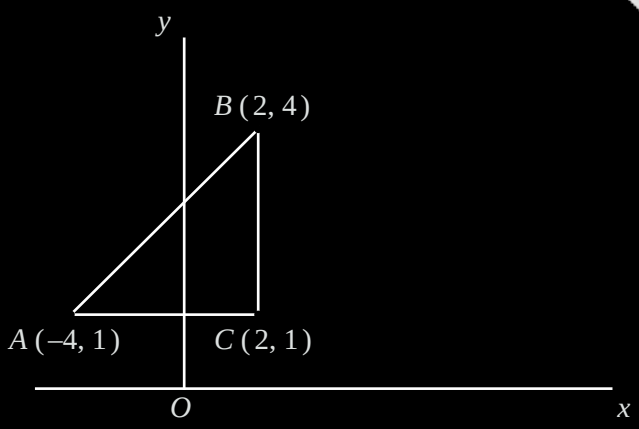
$b =$ .....

$c =$ .....[2]

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Triangle ABC is translated by $\begin{pmatrix} 9 \\ 2 \end{pmatrix}$ onto triangle $A'B'C'$.

(a) Find the coordinates of C'.

Answer (..... , ...) [1]

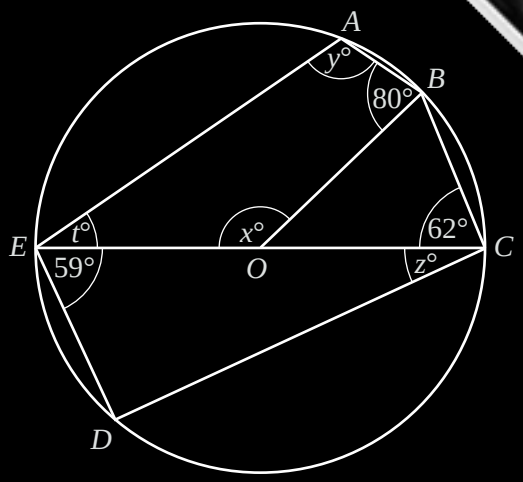
(b) What special type of quadrilateral is $BCC'B'$?

Answer ..... [1]

(c) Find the area of quadrilateral $BCC'B'$.

Answer ..... units^2 [2]

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Find

(a) x,

Answer $x =$ .....[1]

(b) y,

Answer $y =$ .....[1]

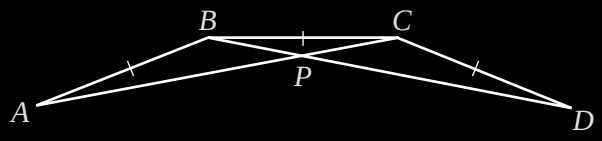
(c) z,

Answer $z =$ .....[1]

(d) t.

Answer $t =$ .....[1]

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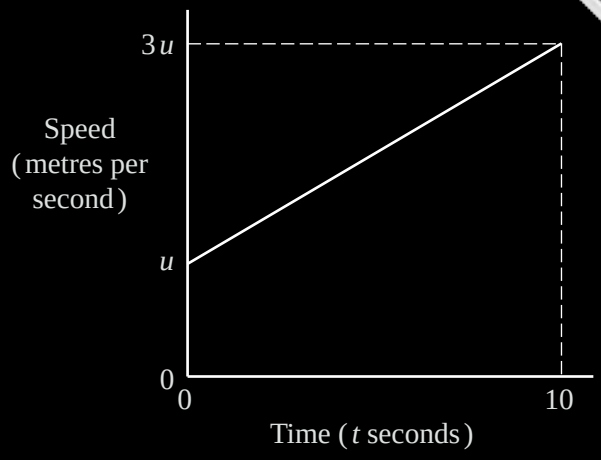
(i) Calculate $B\hat{C}A$.

Answer .....[1]

(ii) Calculate $D\hat{P}C$.

Answer .....[1]

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(a) Find an expression, in terms of u, for the acceleration.

Answer ..... m/s^2 [1]

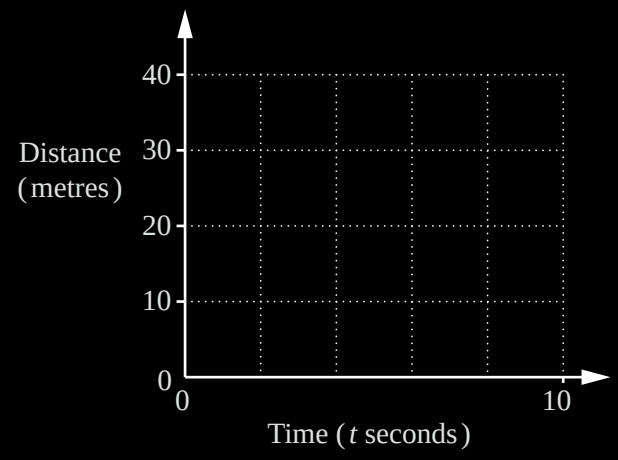
(b) The distance travelled by the cyclist from $t = 0$ to $t = 10$ is 40 m.

(i) Find the value of u.

Answer $u =$ .....[2]

(ii) On the grid below, sketch the distance-time graph of the cyclist.

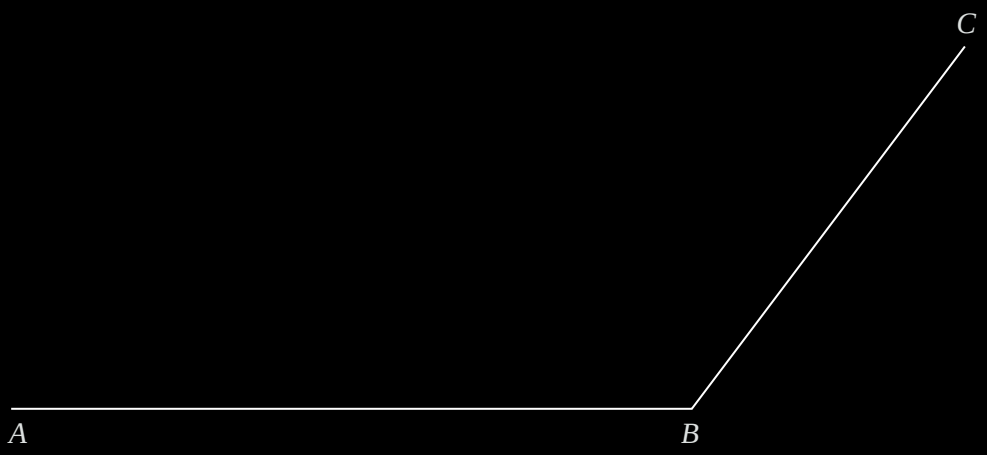
Answer



[2]

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